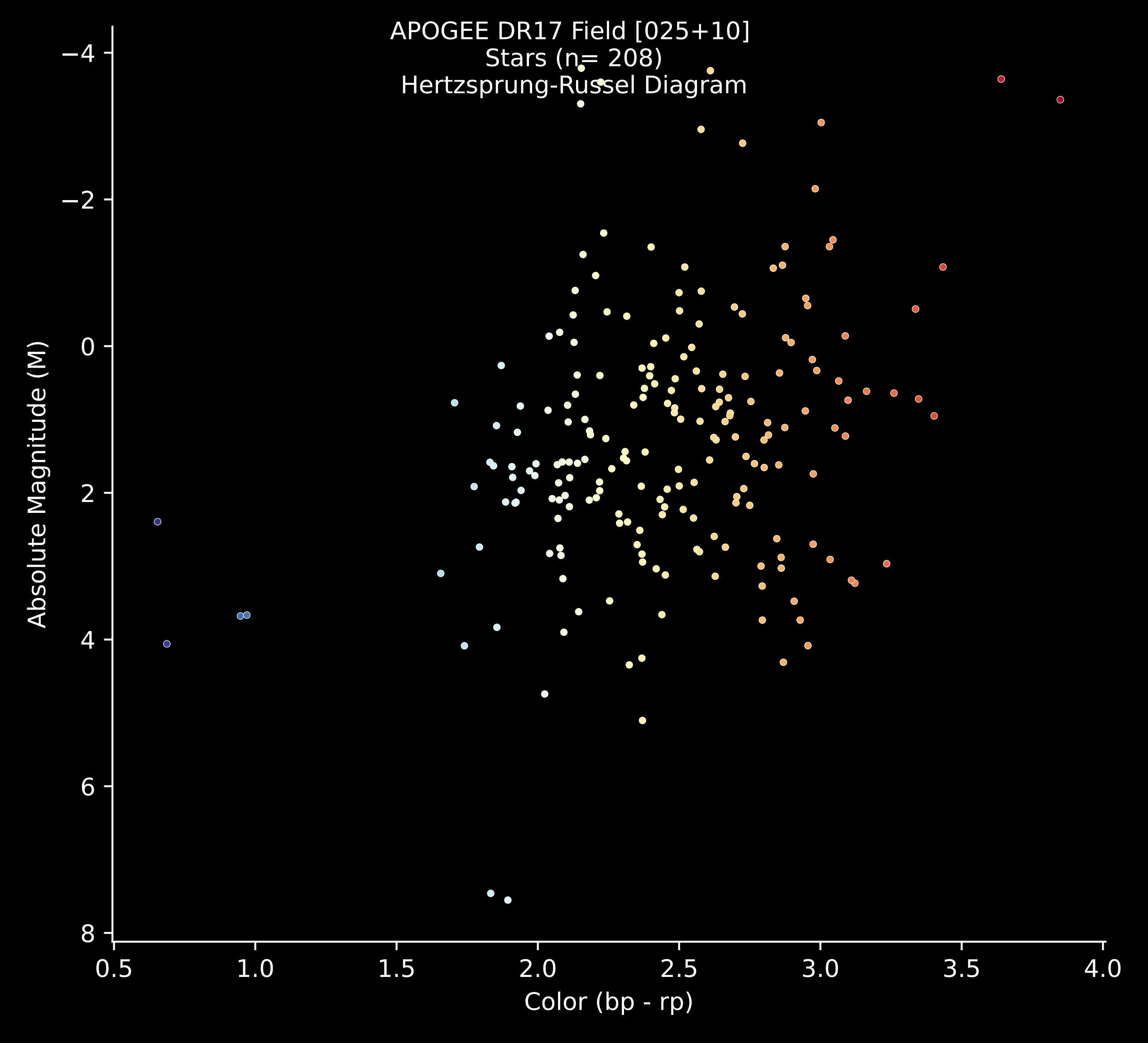
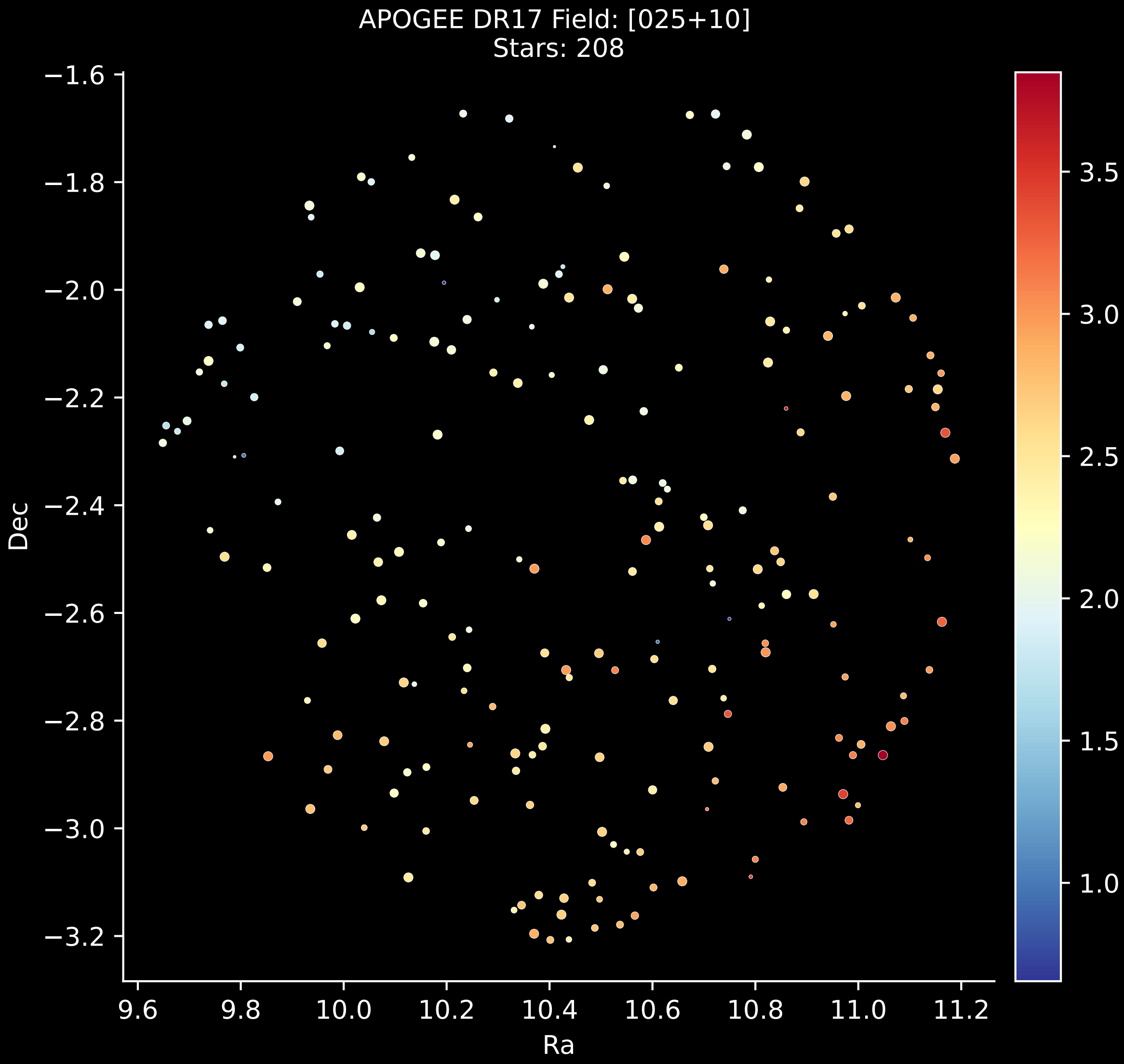
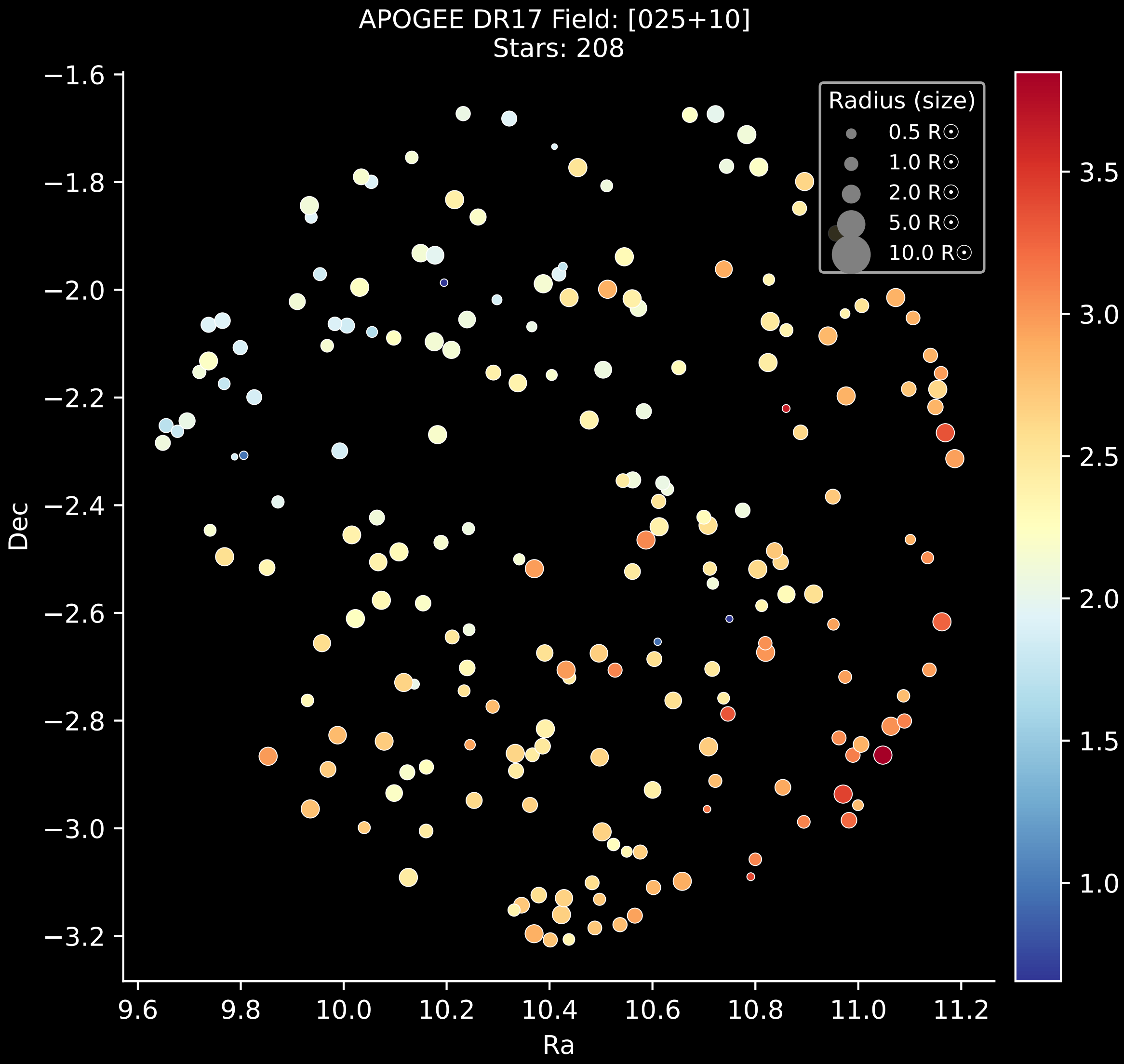
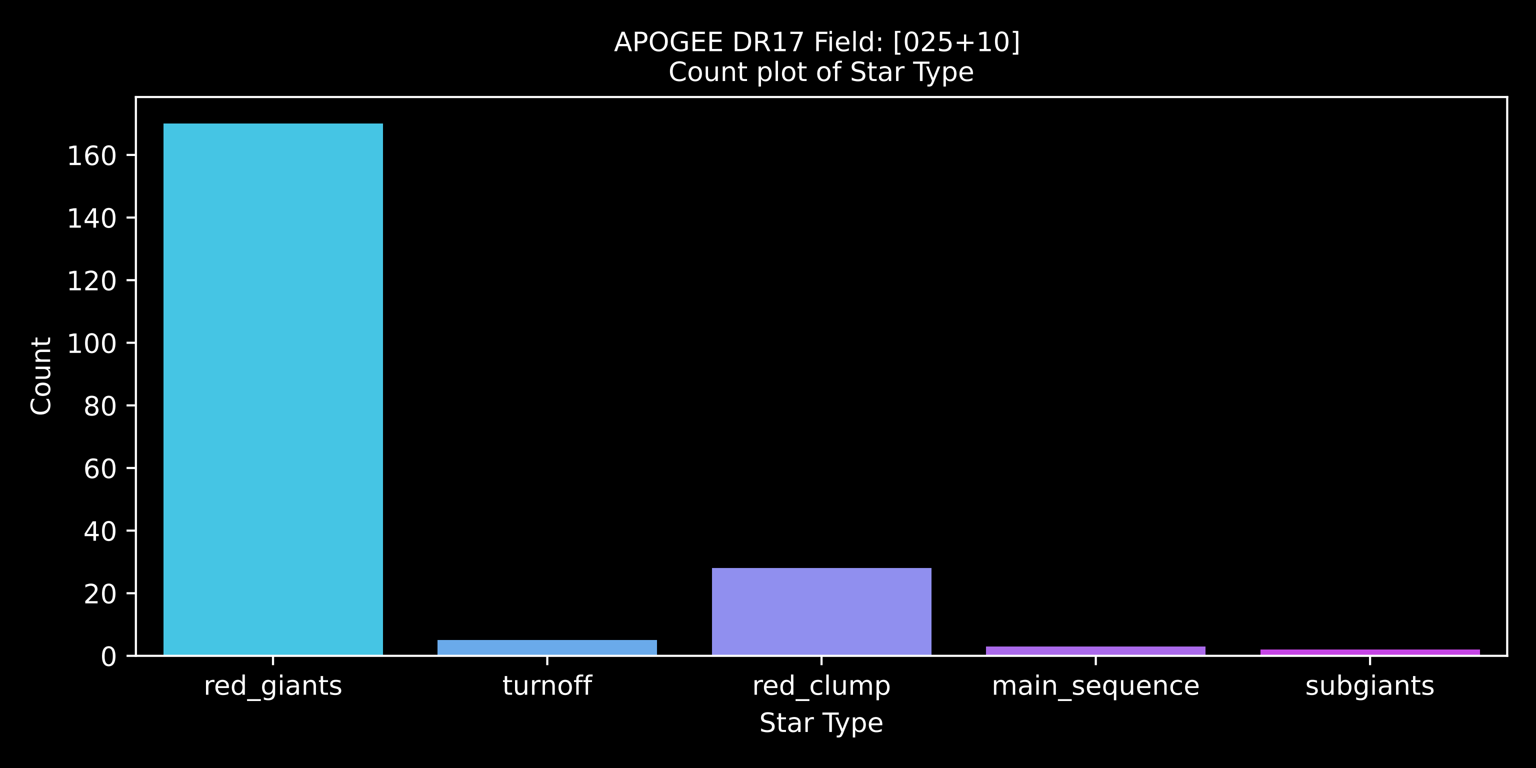
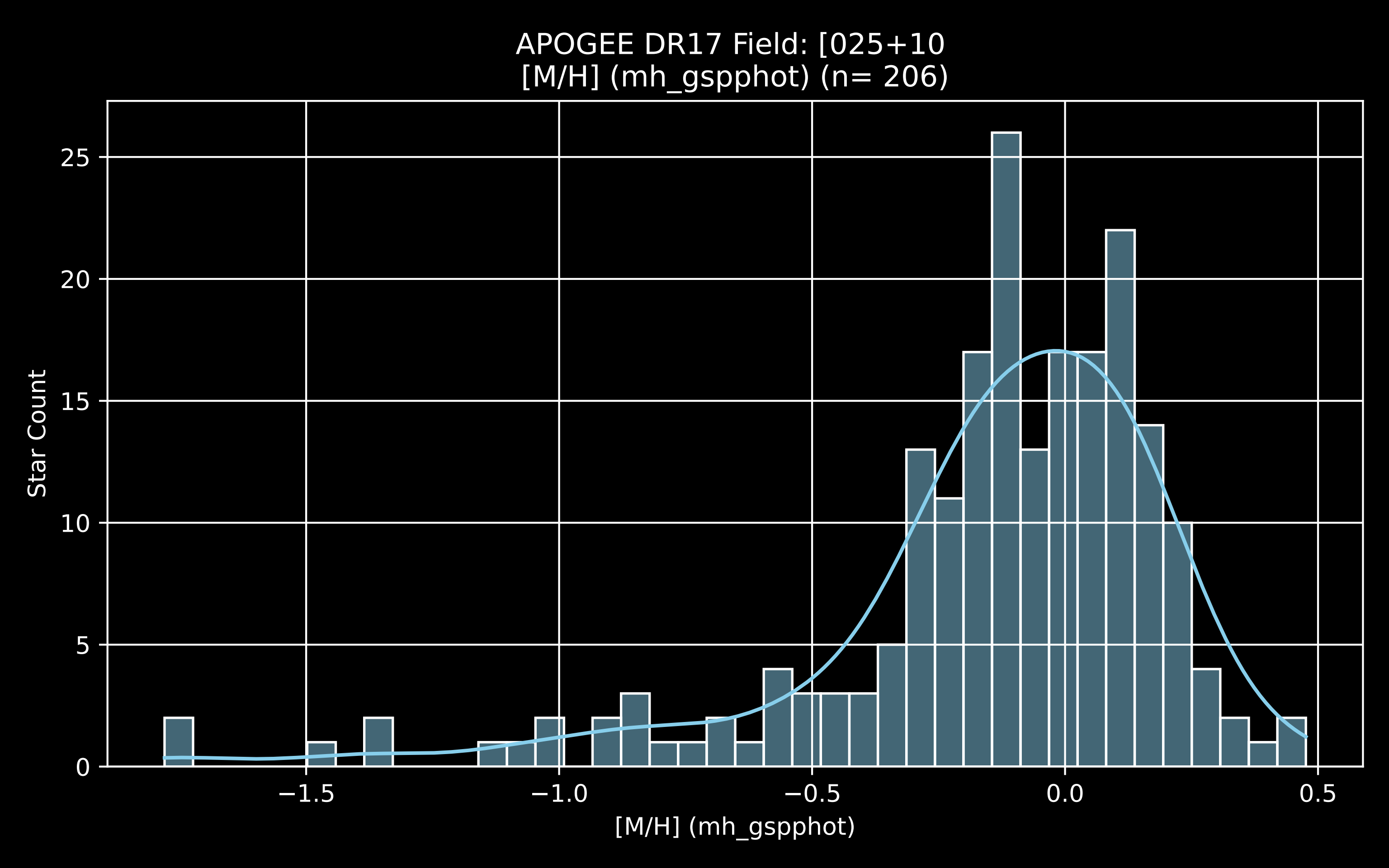
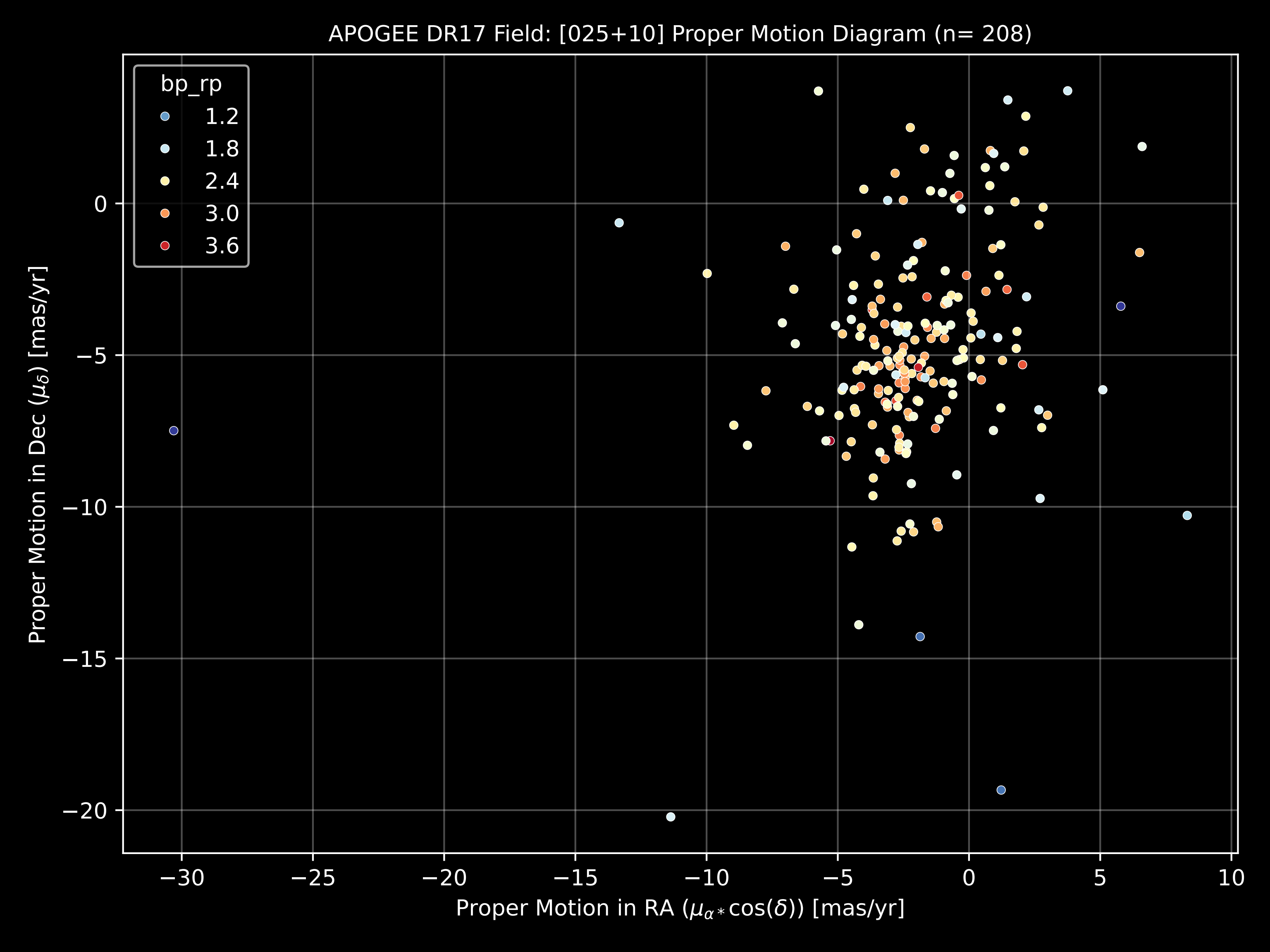

APOGEE DR17 Field [025+10]
Stars (n= 208)
Hertzsprung-Russel Diagram
Absolute Magnitude (M)
Color (bp - rp)
−4
−2
0
2
4
6
8
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0

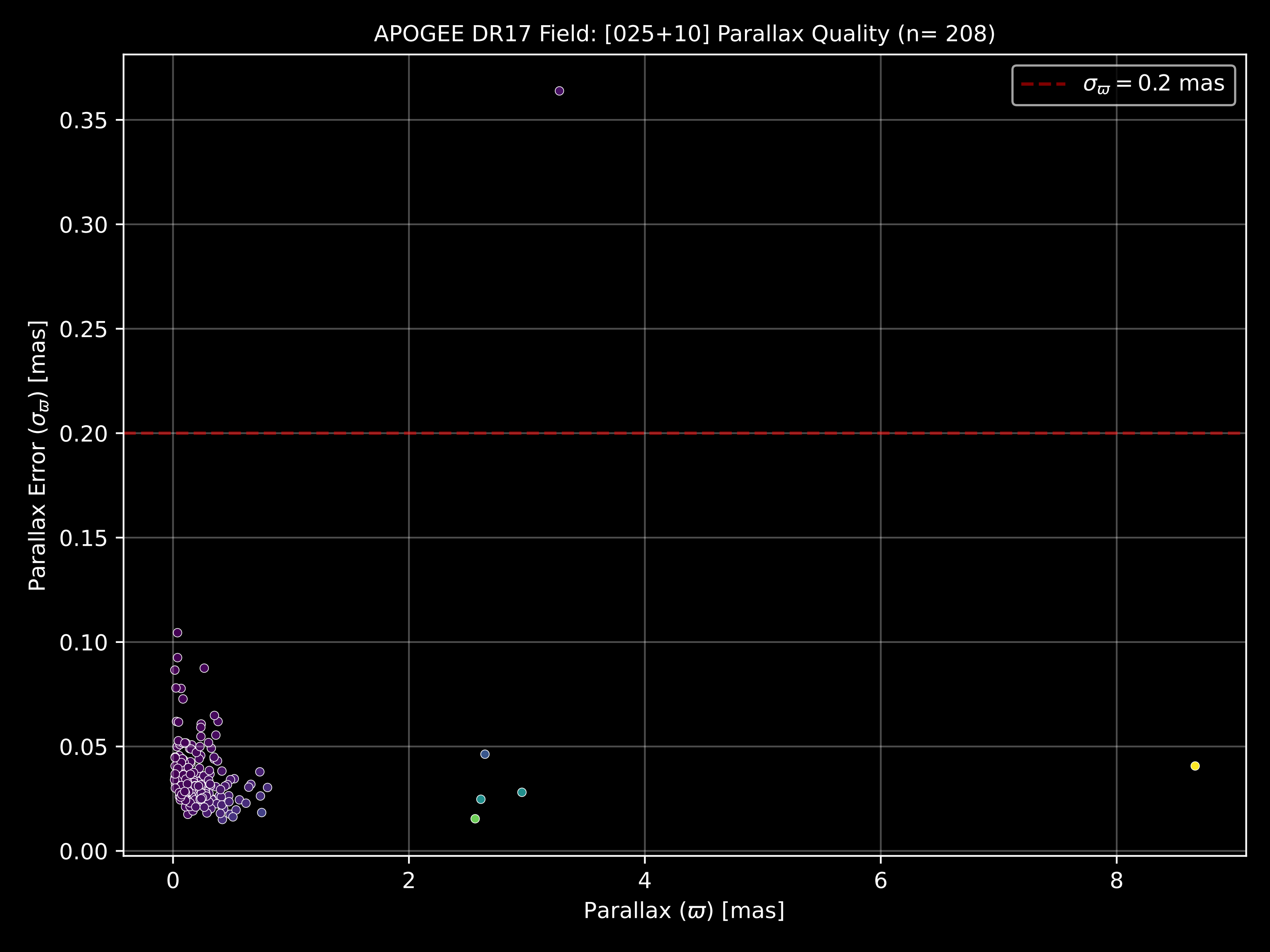

APOGEE DR17 Field: [025+10]
Stars: 208
−1.6
−1.8
−2.0
−2.2
−2.4
−2.6
−2.8
−3.0
−3.2
Dec
9.6
9.8
10.0
10.2
10.4
10.6
10.8
11.0
11.2
Ra
3.5
3.0
2.5
2.0
1.5
1.0
bp_rp


APOGEE DR17 Field: [025+10]
Stars: 208
Radius (size)
0.5 R☉
1.0 R☉
2.0 R☉
5.0 R☉
10.0 R☉
Dec
Ra
bp_rp (color)
−1.6
−1.8
−2.0
−2.2
−2.4
−2.6
−2.8
−3.0
−3.2
9.6
9.8
10.0
10.2
10.4
10.6
10.8
11.0
11.2
3.5
3.0
2.5
2.0
1.5
1.0

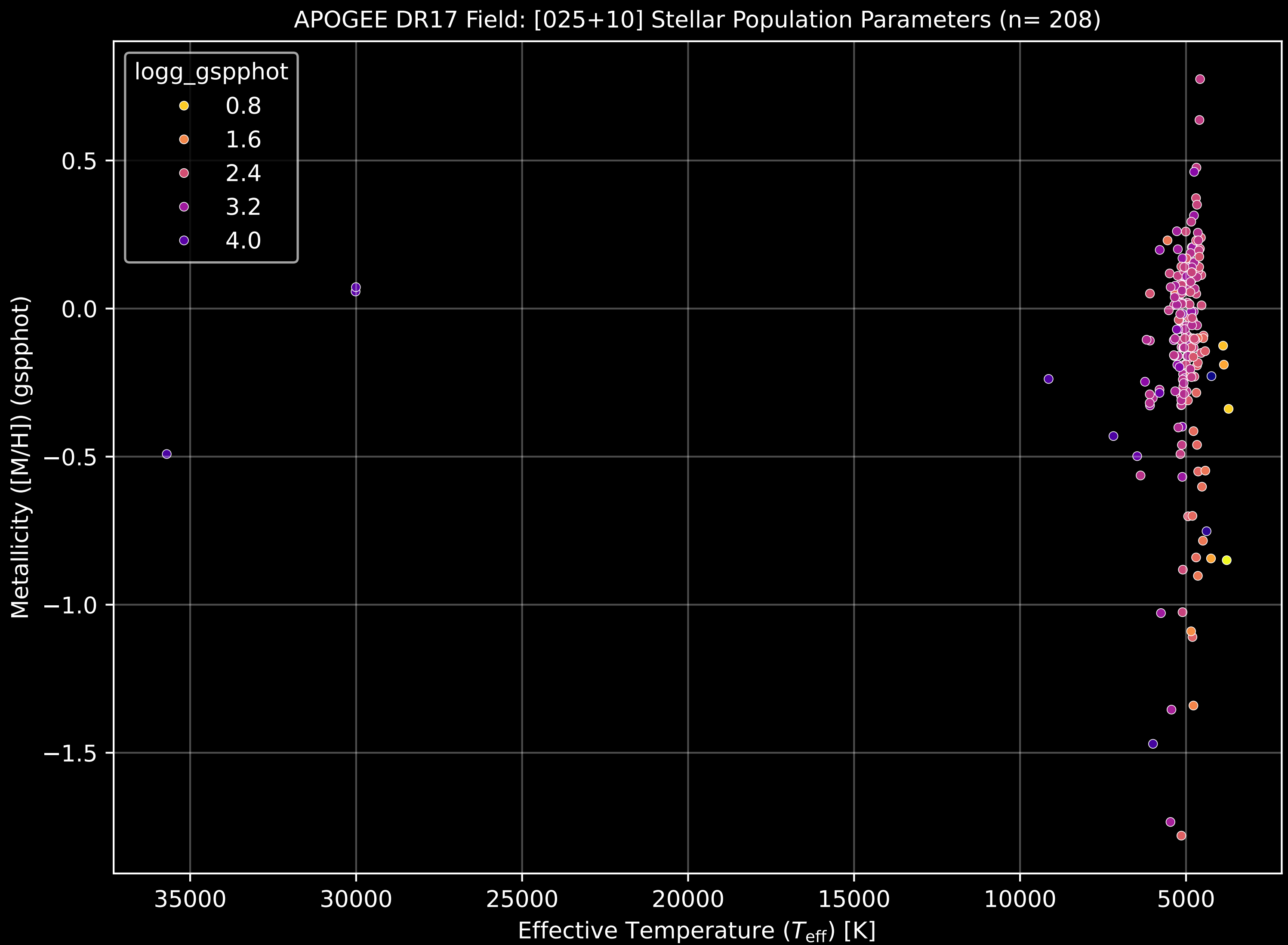

APOGEE DR17 Field: [025+10]
Count plot of Star Type
Count
0
20
40
60
80
100
120
140
160
red_giants
turnoff
red_clump
main_sequence
subgiants
Star Type


APOGEE DR17 Field: [025+10
[M/H] (mh_gspphot) (n= 206)
Star Count
[M/H] (mh_gspphot)
0
5
10
15
20
25
−1.5
−1.0
−0.5
0.0
0.5

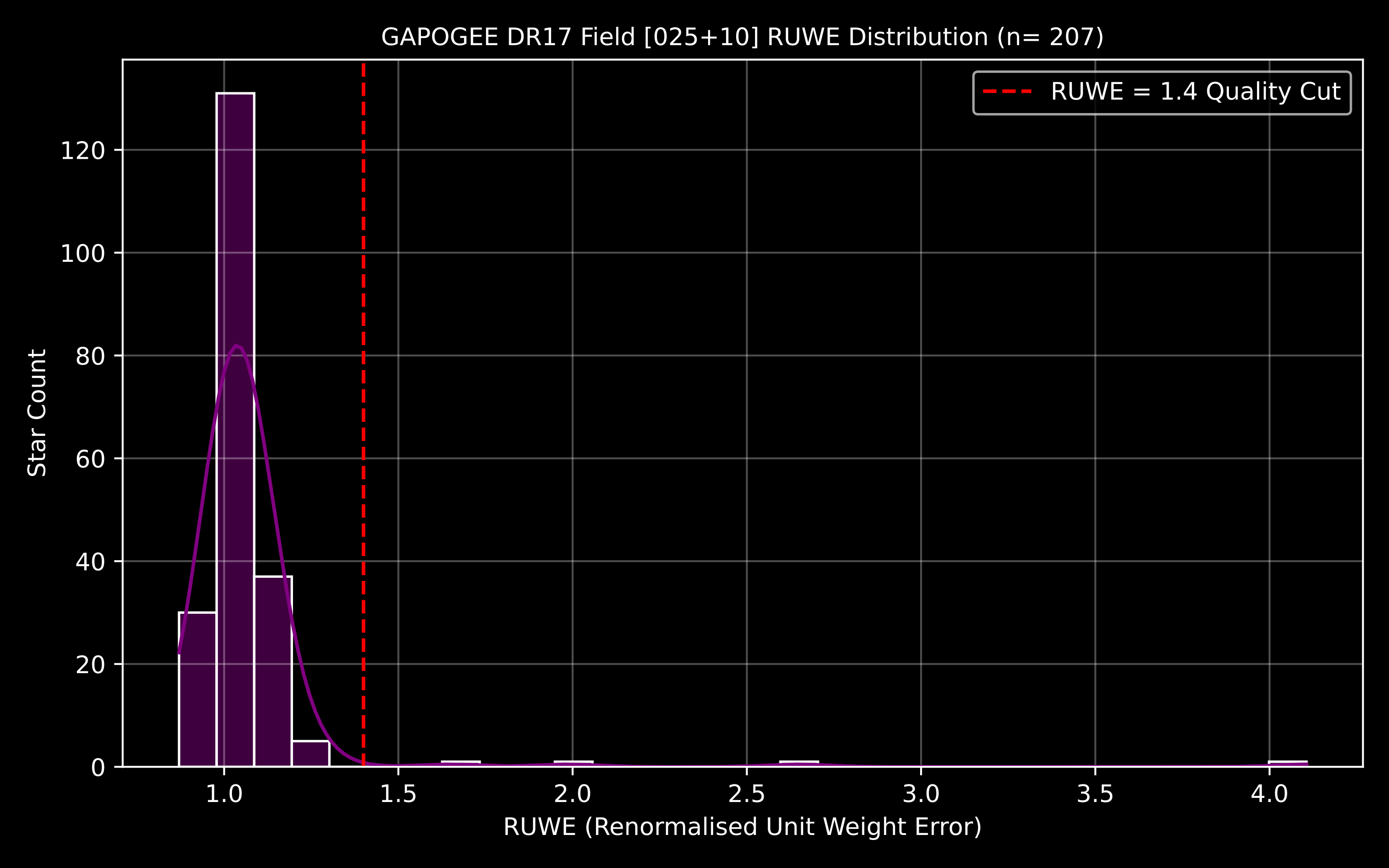

APOGEE DR17 Field: [025+10] Proper Motion Diagram (n= 208)
bp_rp
1.2
1.8
2.4
3.0
3.6
Proper Motion in Dec (μ_δ) [mas/yr]
Proper Motion in RA ($\mu_{\alpha*}\cos(\delta)$) [mas/yr]
0
−5
−10
−15
−20
−30
−25
−20
−15
−10
−5
0
5
10

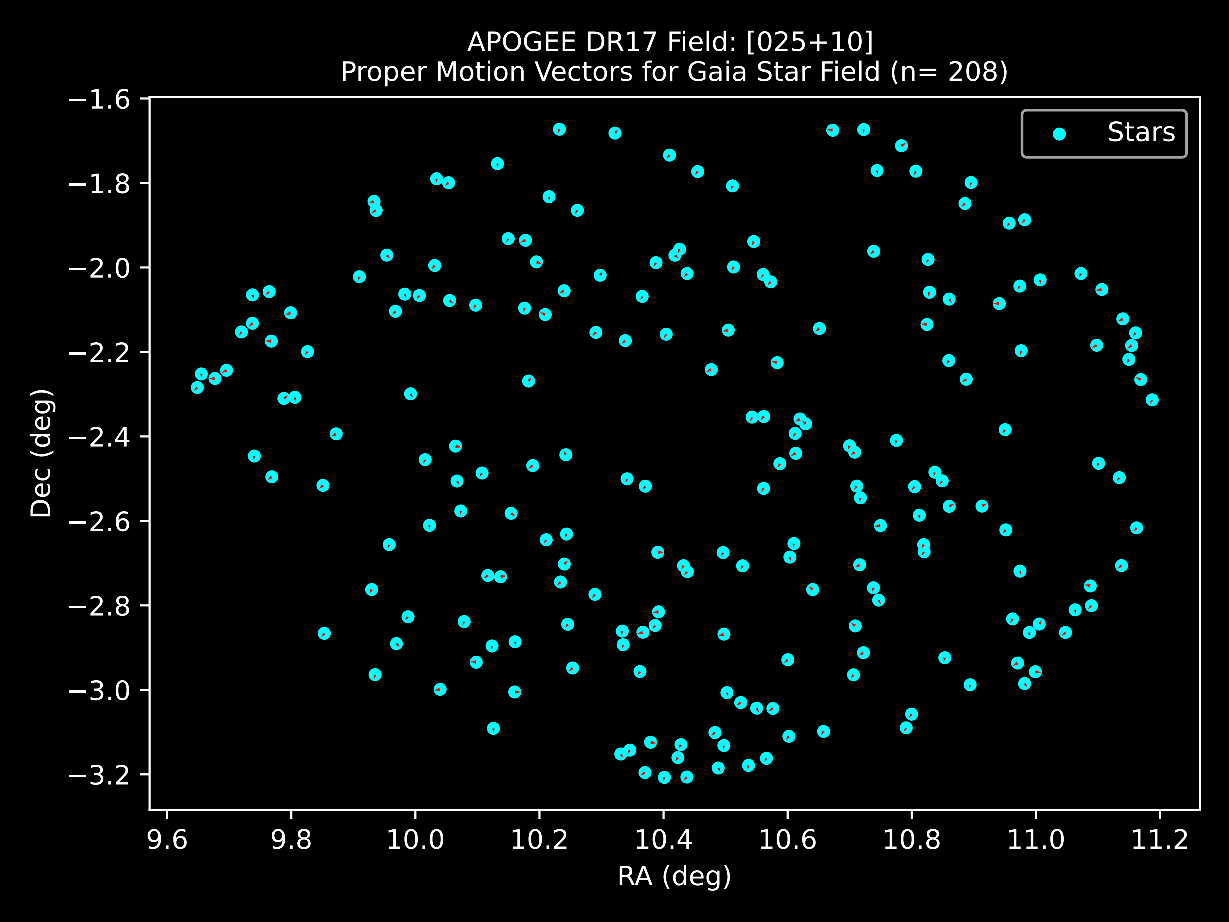


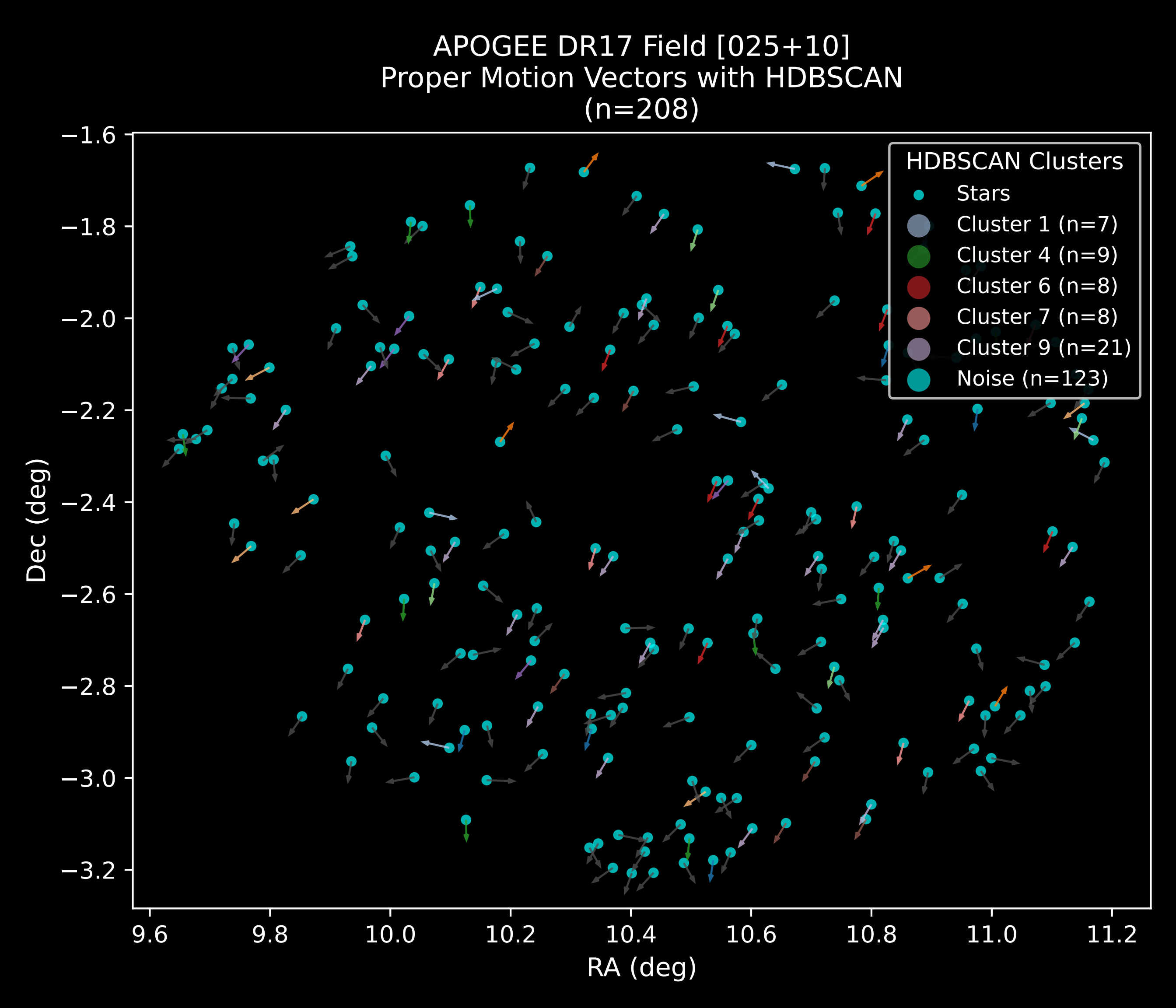
APOGEE DR17 Field: [025+10] Parallax Quality (n= 208)
$\sigma_\varpi = 0.2$ mas
Parallax Error (σ_ϖ) [mas]
Parallax (ϖ) [mas]
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0
2
4
6
8

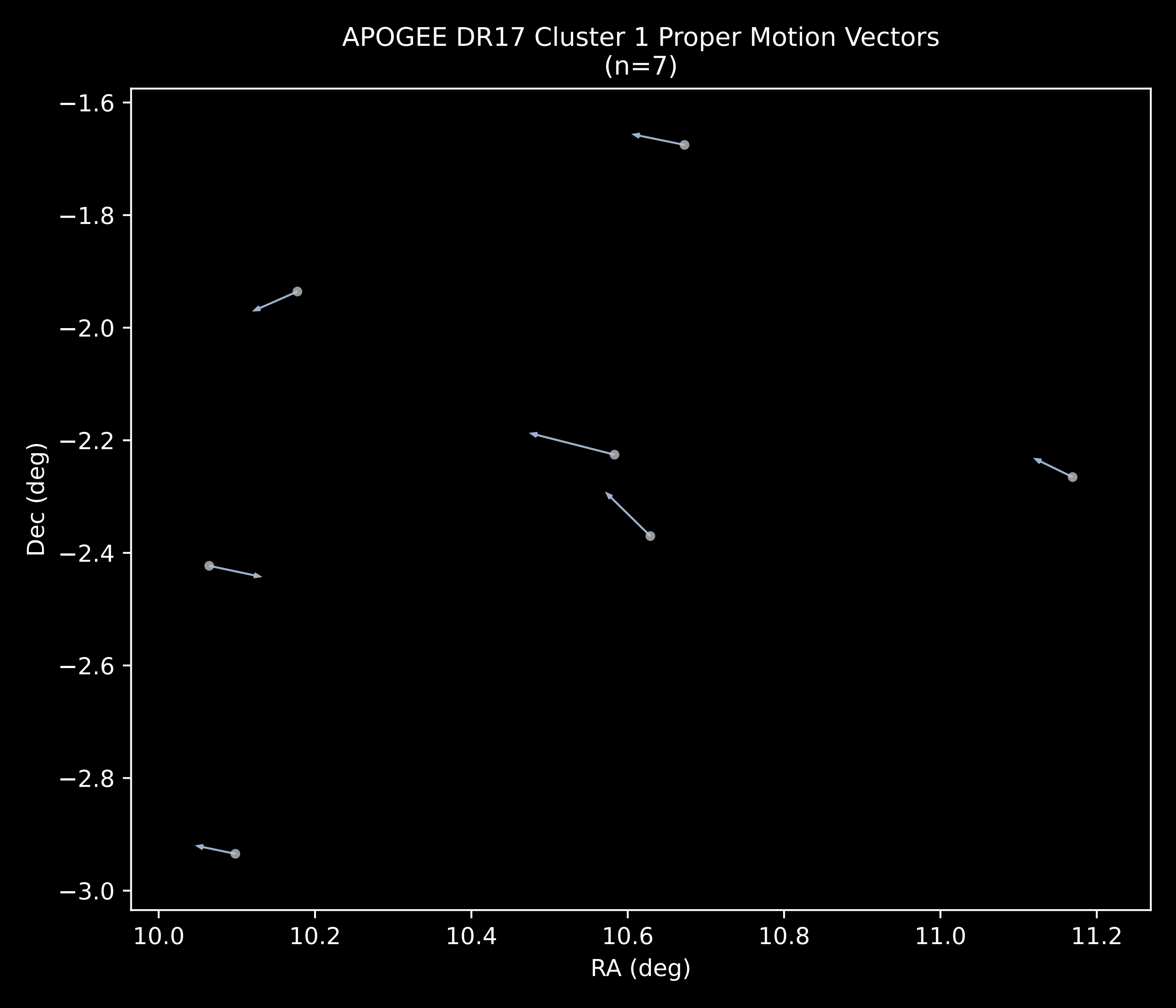


APOGEE DR17 Field: [025+10] Stellar Population Parameters (n= 208)
logg_gspphot
0.8
1.6
2.4
3.2
4.0
Metallicity ([M/H]) (gspphot)
0.5
0.0
−0.5
−1.0
−1.5
35000
30000
25000
20000
15000
10000
5000
Effective Temperature (T_{eff}) [K]

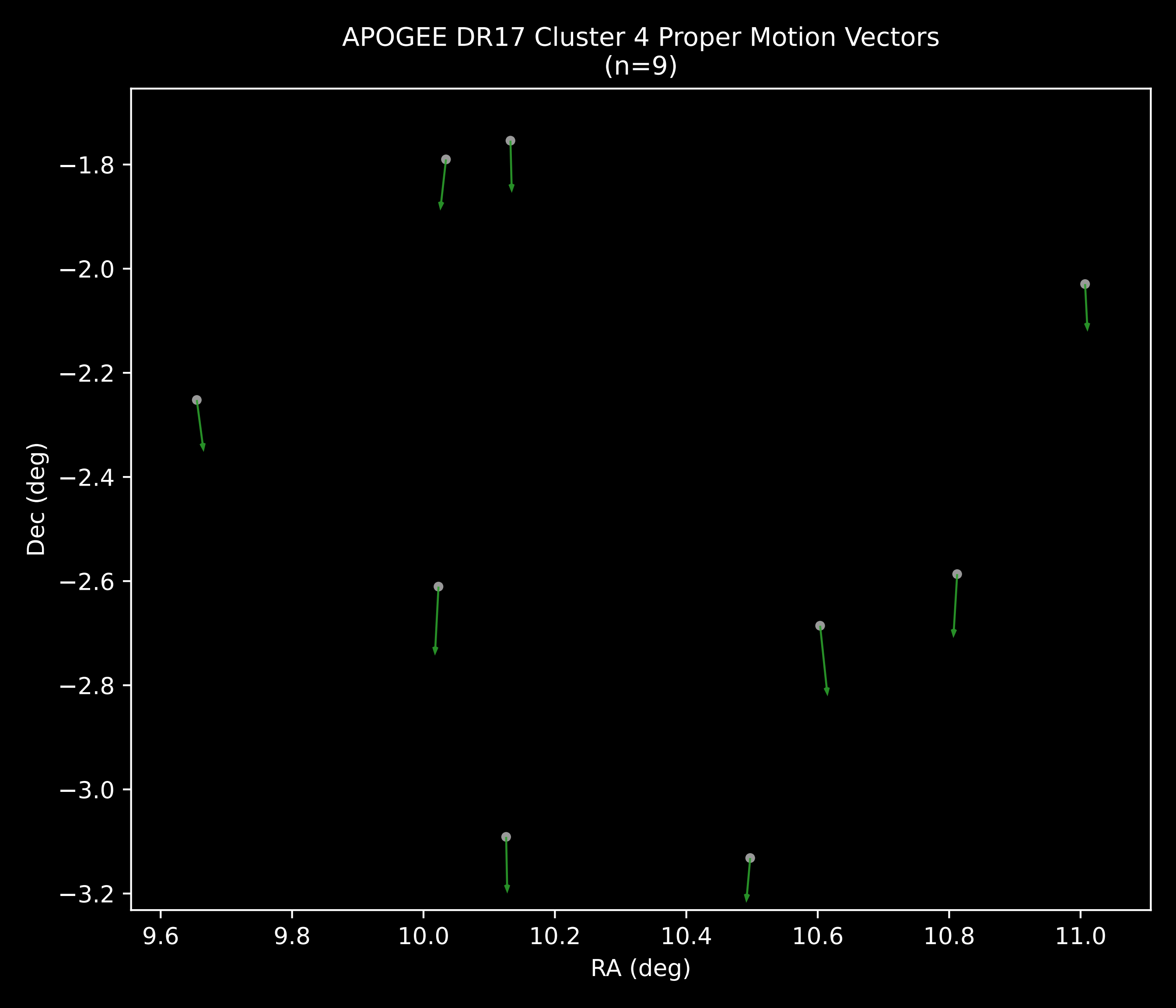

GAPOGEE DR17 Field [025+10] RUWE Distribution (n= 207)
RUWE = 1.4 Quality Cut
Star Count
0
20
40
60
80
100
120
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
RUWE (Renormalised Unit Weight Error)


APOGEE DR17 Field: [025+10]
Proper Motion Vectors for Gaia Star Field (n= 208)
Stars
Dec (deg)
RA (deg)
−1.6
−1.8
−2.0
−2.2
−2.4
−2.6
−2.8
−3.0
−3.2
9.6
9.8
10.0
10.2
10.4
10.6
10.8
11.0
11.2

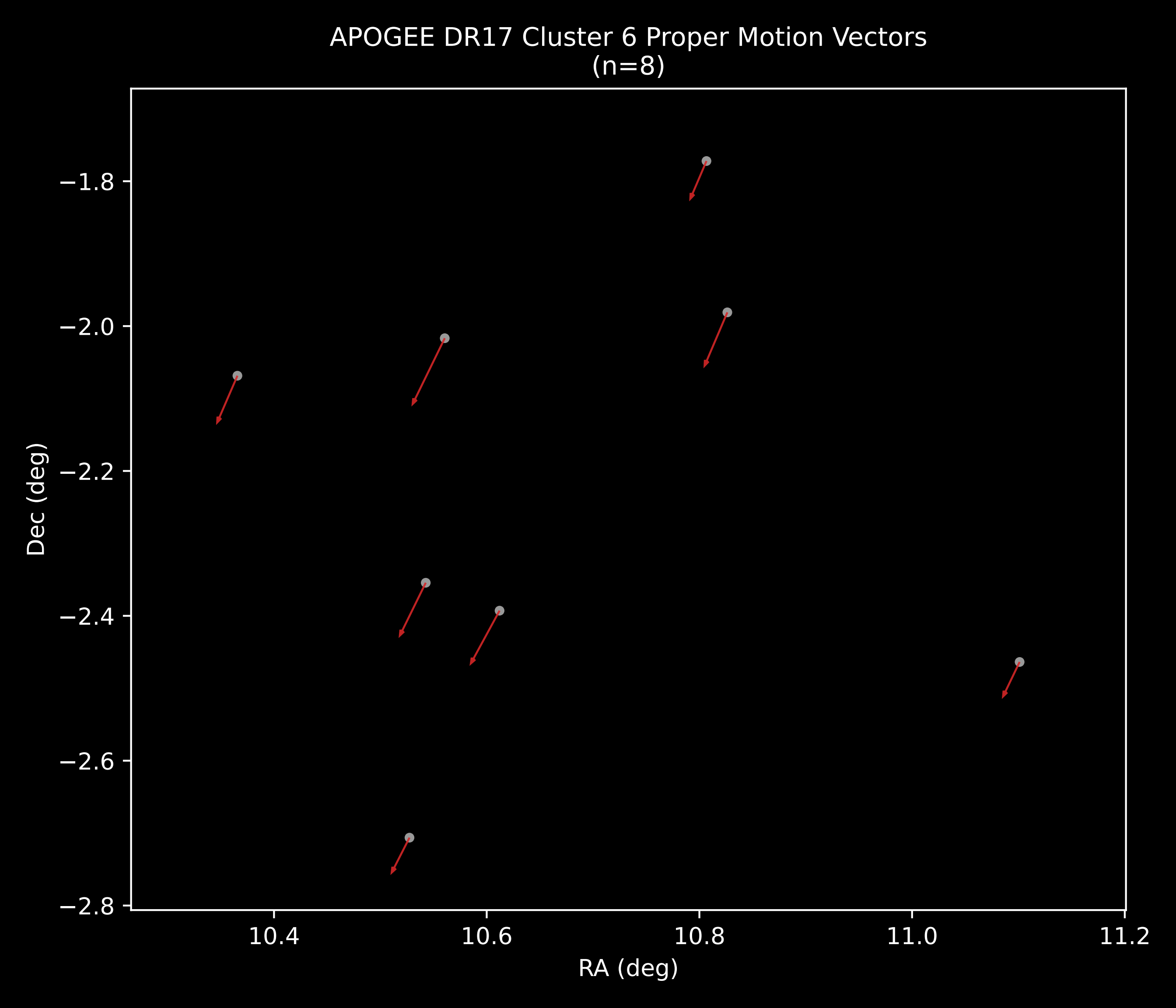

APOGEE DR17 Field [025+10]
Proper Motion Vectors with HDBSCAN
(n=208)
HDBSCAN Clusters
Stars
Cluster 1 (n=7)
Cluster 4 (n=9)
Cluster 6 (n=8)
Cluster 7 (n=8)
Cluster 9 (n=21)
Noise (n=123)
Dec (deg)
RA (deg)
−1.6
−1.8
−2.0
−2.2
−2.4
−2.6
−2.8
−3.0
−3.2
9.6
9.8
10.0
10.2
10.4
10.6
10.8
11.0
11.2

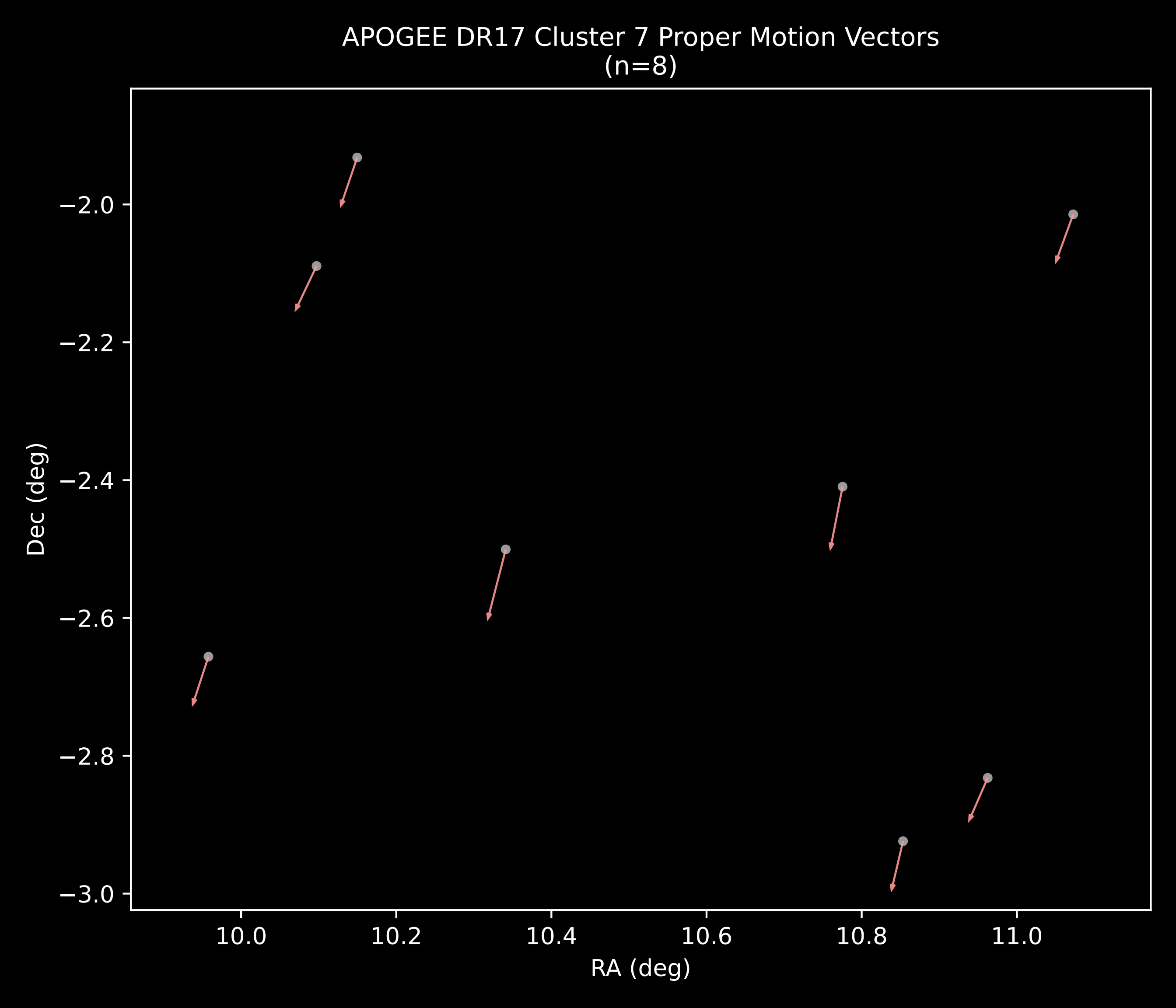


APOGEE DR17 Cluster 1 Proper Motion Vectors
(n=7)
−1.6
−1.8
−2.0
−2.2
−2.4
−2.6
−2.8
−3.0
Dec (deg)
10.0
10.2
10.4
10.6
10.8
11.0
11.2
RA (deg)

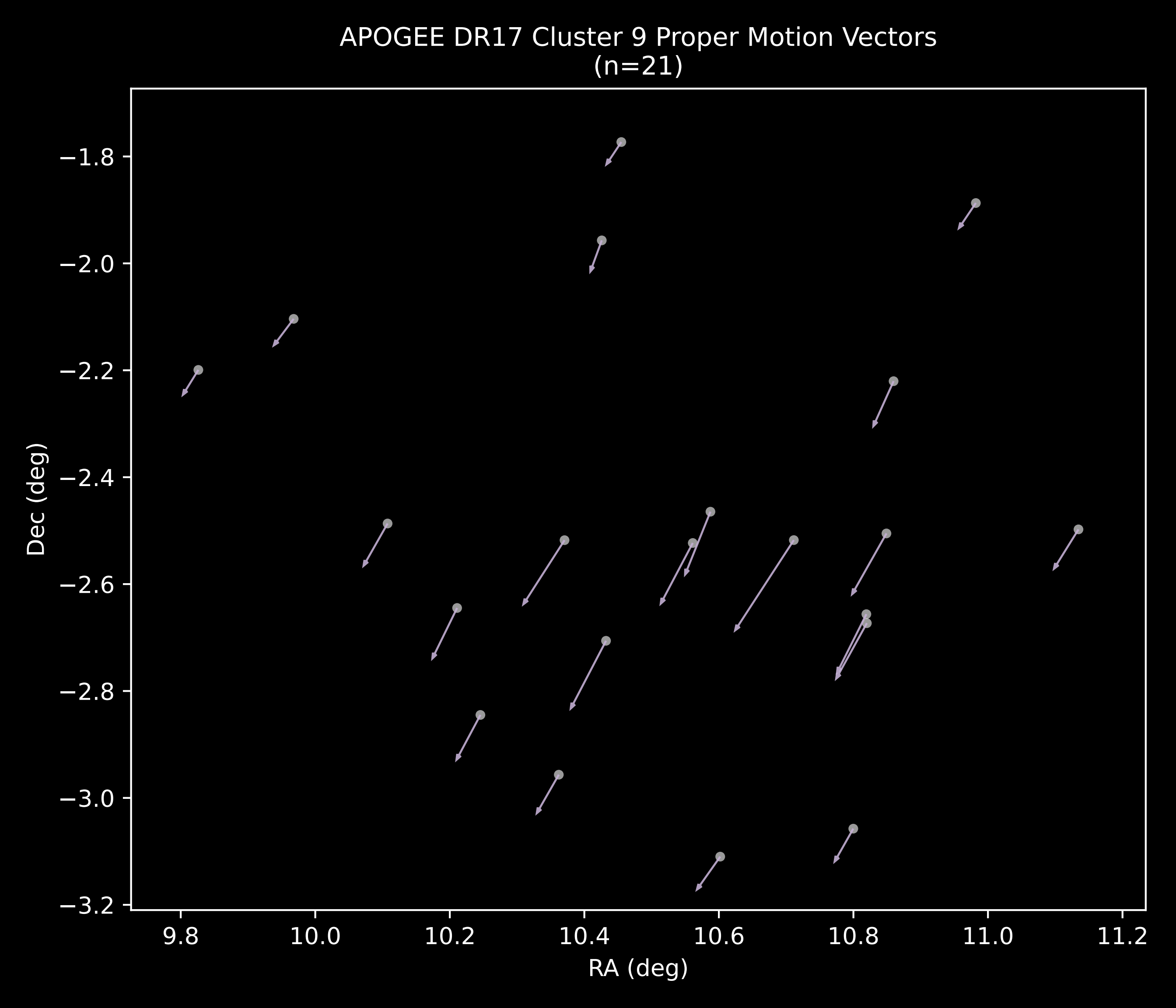


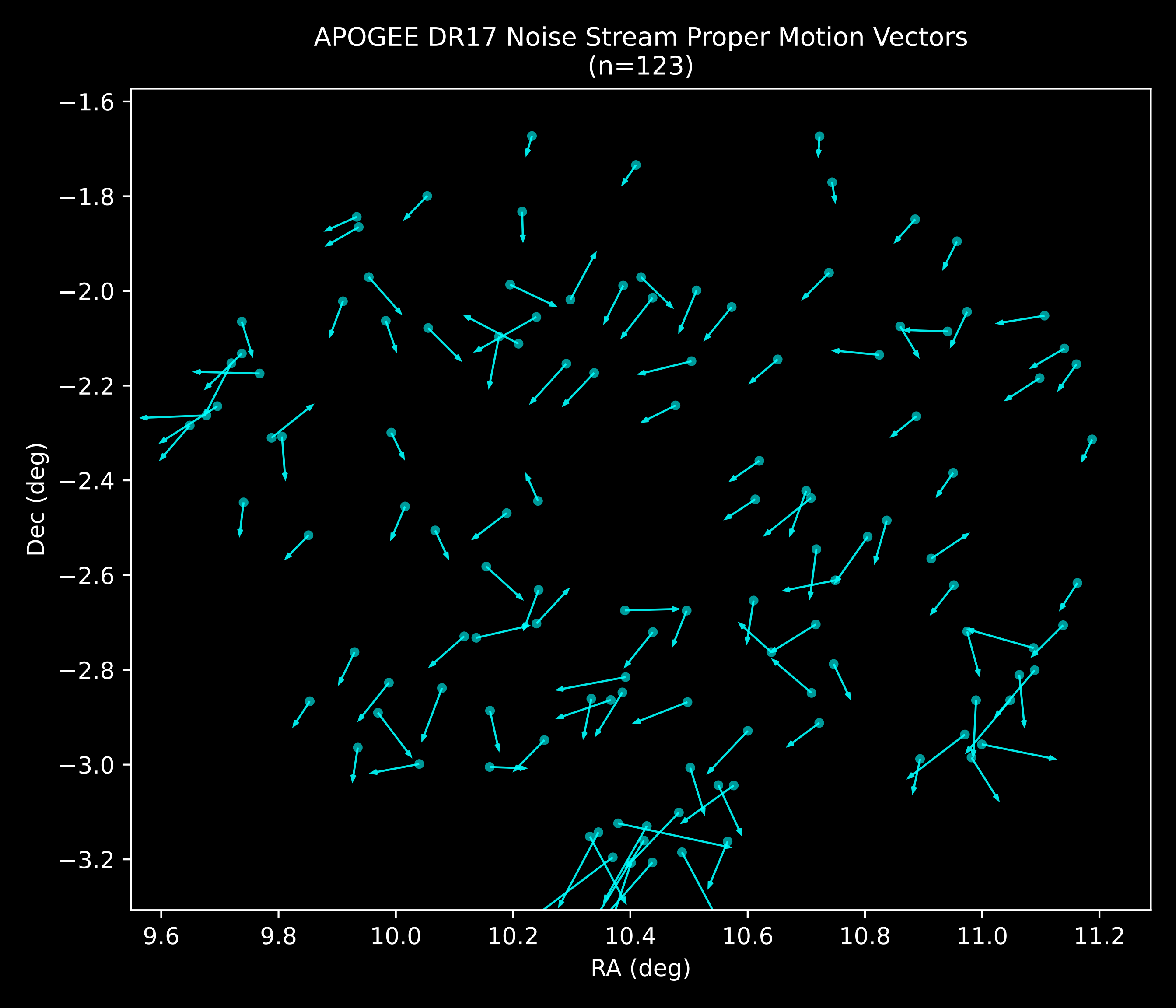
APOGEE DR17 Cluster 4 Proper Motion Vectors
(n=9)
Dec (deg)
RA (deg)
−1.8
−2.0
−2.2
−2.4
−2.6
−2.8
−3.0
−3.2
9.6
9.8
10.0
10.2
10.4
10.6
10.8
11.0


APOGEE DR17 Cluster 6 Proper Motion Vectors
(n=8)
−1.8
−2.0
−2.2
−2.4
−2.6
−2.8
Dec (deg)
10.4
10.6
10.8
11.0
11.2
RA (deg)



APOGEE DR17 Cluster 7 Proper Motion Vectors
(n=8)
Dec (deg)
−2.0
−2.2
−2.4
−2.6
−2.8
−3.0
RA (deg)
10.0
10.2
10.4
10.6
10.8
11.0


APOGEE DR17 Cluster 9 Proper Motion Vectors
(n=21)
Dec (deg)
RA (deg)
−1.8
−2.0
−2.2
−2.4
−2.6
−2.8
−3.0
−3.2
9.8
10.0
10.2
10.4
10.6
10.8
11.0
11.2


APOGEE DR17 Noise Stream Proper Motion Vectors
(n=123)
Dec (deg)
RA (deg)
−1.6
−1.8
−2.0
−2.2
−2.4
−2.6
−2.8
−3.0
−3.2
9.6
9.8
10.0
10.2
10.4
10.6
10.8
11.0
11.2